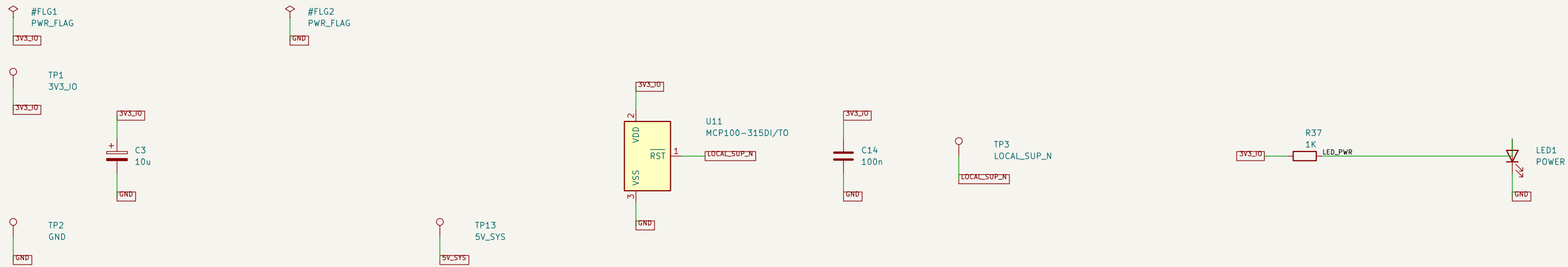
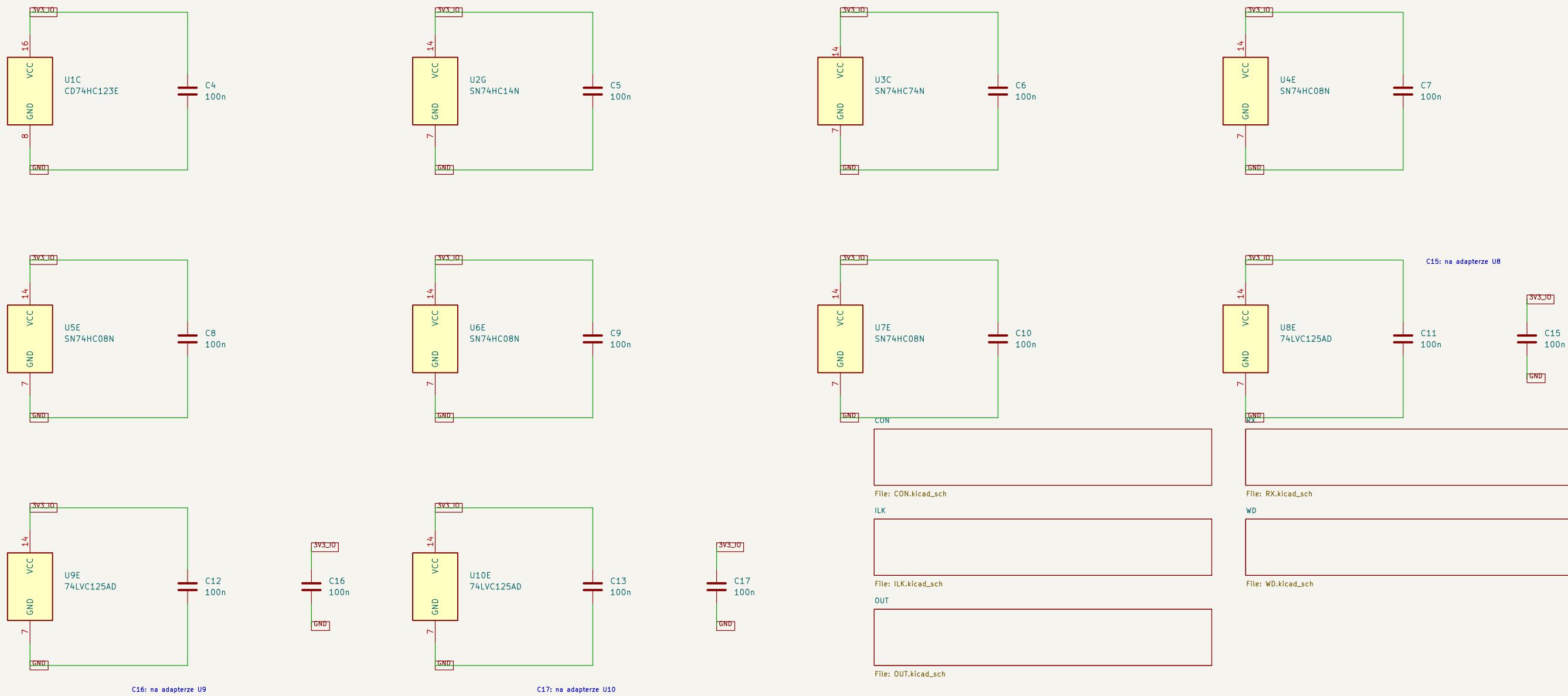


P04-R1 / 01 ZASILANIE I ODSPRZEGANIE



Zasilanie 3V3_I0 z P02. 5V_SYS z LV04 jest dostępne na J1.1 i TP13; nie zasila logiki.



U11: MCP100 w bondout D, 1 RESET / 2 VDD / 3 GND. Lokalny reset nie jest podłączony push-pull do SAFE_N.

Hardware interlock, watchdog and manual arm; review, hardware not tested

Prototype / schematic + PCB review

Sheet: /

File: P04.kicad_sch

Title: EGRLab P04 SAFE

Size: A3

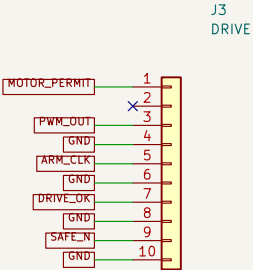
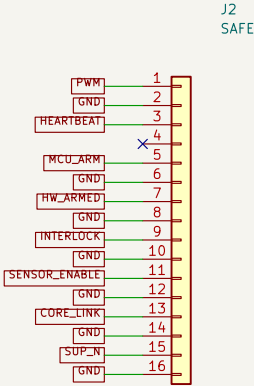
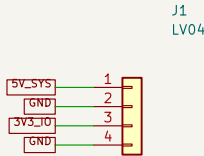
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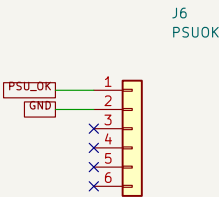
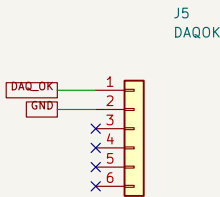
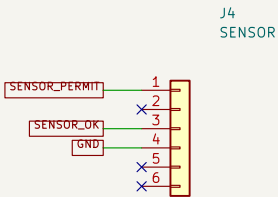
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Id: 1/6

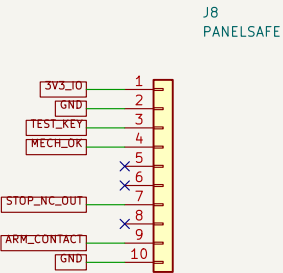
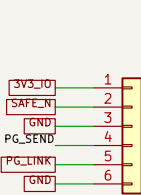
J1 LV04: przewody AWG22 200 mm. J2 SAFE: tasma 16 zyl AWG28 150 mm. PTH + kotwa 12 mm.



Klucze: SAFE 4 / DRIVE 2 / SENSOR 2 / DAQOK 4 / PSUOK 5. Pad klucza NC; usuniety styk w gniezdzie.



SENSOR M2.2: IDC 6p; 1 permit, 2 KEY, 3 ready, 4 GND, 5/6 NC. Wymagana zgodna wiazka P08.



PANELSAFE: STOP NC miedzy 1 i 7. ARM NO miedzy 9 i 10. Nie laczyc szyny P00 3V3 z P04 3V3_IO.

Hardware interlock, watchdog and manual arm; review, hardware not tested

Prototype / schematic + PCB review

Sheet: /CON/

File: CON.kicad_sch

Title: EGRLab P04 SAFE

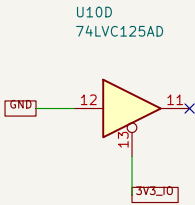
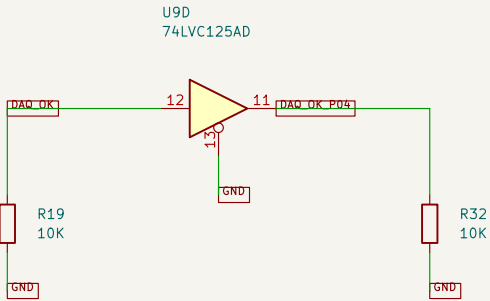
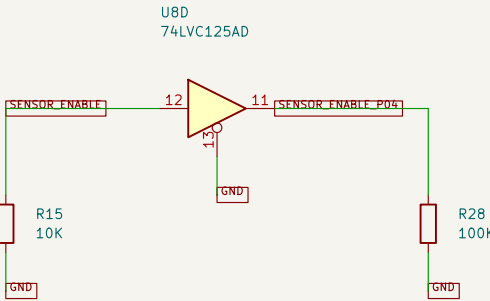
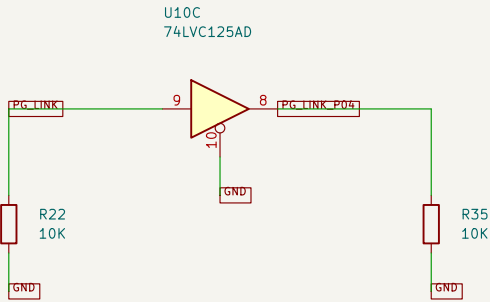
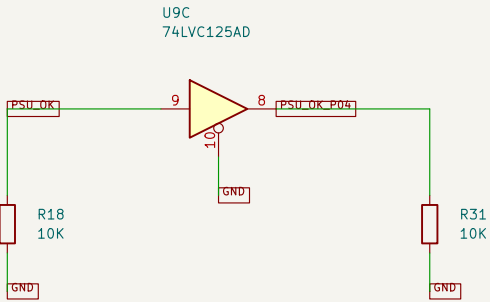
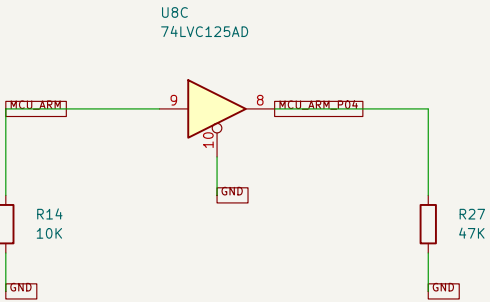
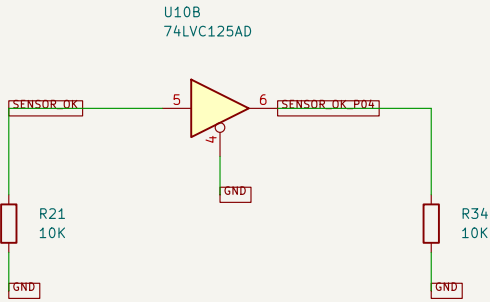
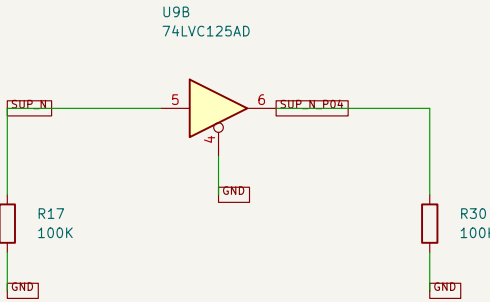
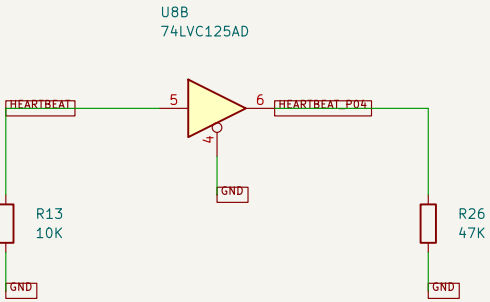
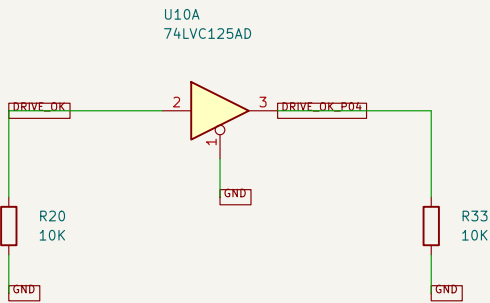
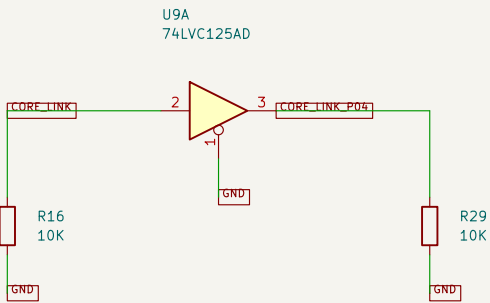
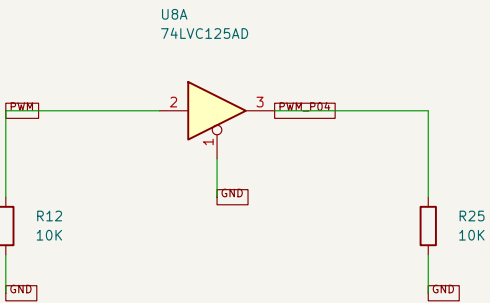
Size: A3

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74LVC125AD Nexperia: loff; pin wejścia nie zasila nieaktywnej domeny. Nie zastapic ukladem HC125.

Hardware interlock, watchdog and manual arm; review, hardware not tested

Prototype / schematic + PCB review

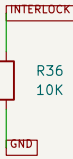
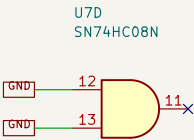
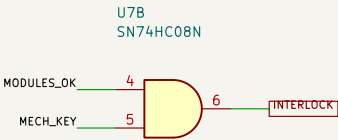
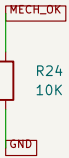
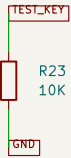
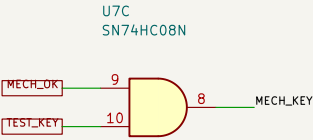
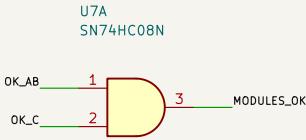
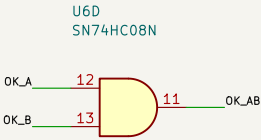
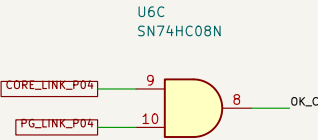
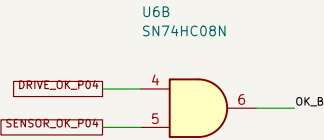
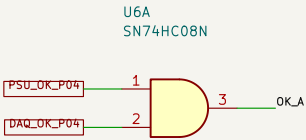
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Title: EGRLab P04 SAFE

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MODULES_OK = PSU_OK & DAQ_OK & DRIVE_OK & SENSOR_OK & CORE_LINK & PG_LINK (po buforach).



INTERLOCK = MODULES_OK & MECH_OK & TEST_KEY. P11 MECH_OK zawiera klucz, oba styki LOGGER i petle TEST.

READY nie zależy od przełącznika, który ma uzbroić. Powrót READY nie wywołuje nowego zbocza ARM.

Hardware interlock, watchdog and manual arm; review, hardware not tested

Prototype / schematic + PCB review

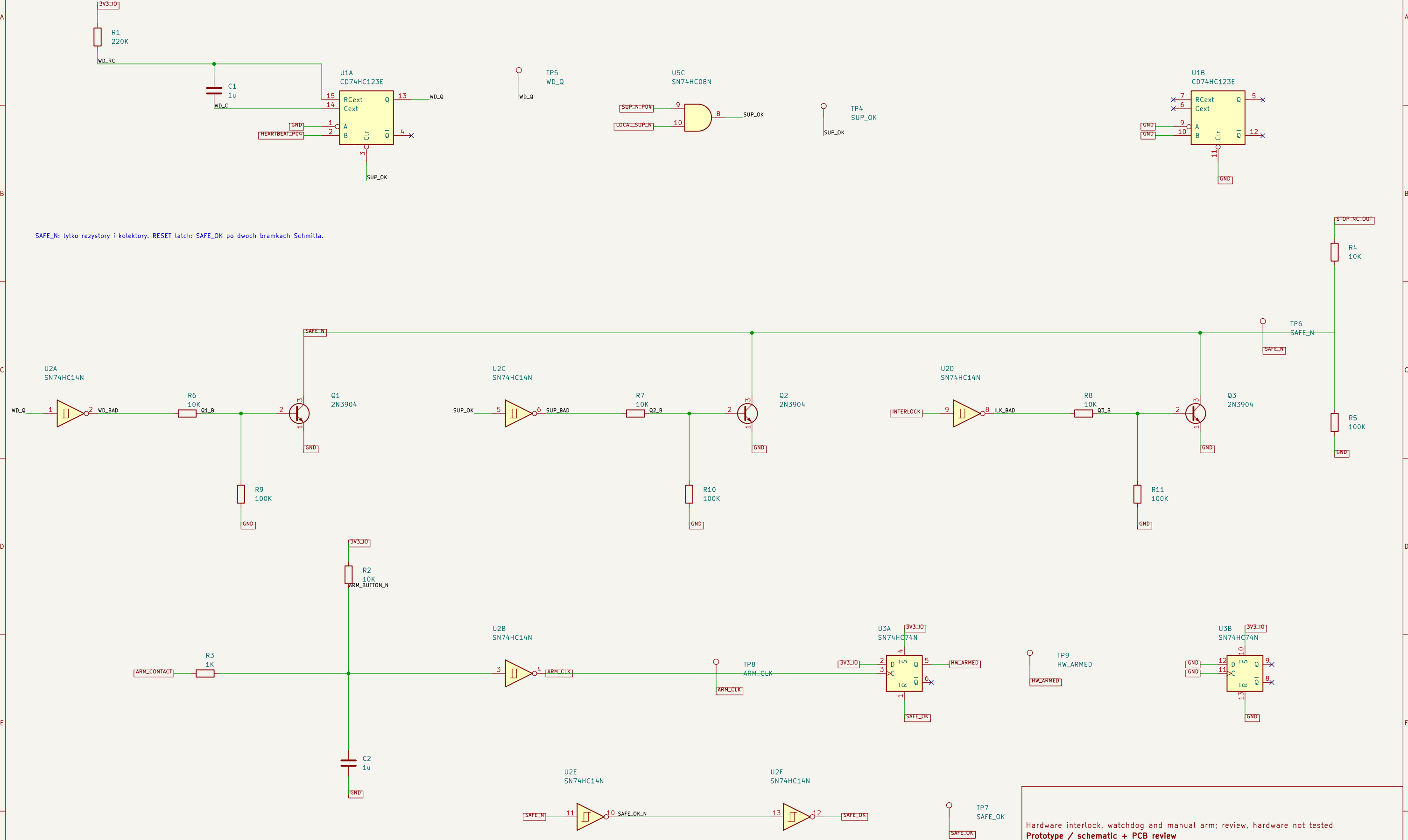
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Title: EGRLab P04 SAFE

Size: A3 Date: 2026-09-25
KiCad E.D.A. 10.0.6

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U1 CLR = SUP_OK, niezależnie od SAFE_N. C1 pomiędzy pinami 15 i 14; pin 14 NIE jest masa.

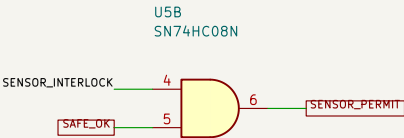
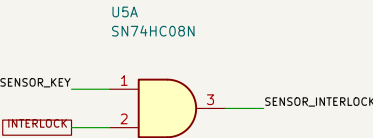
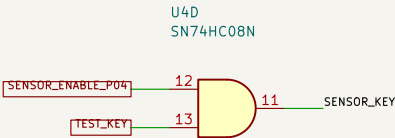
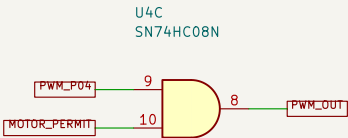
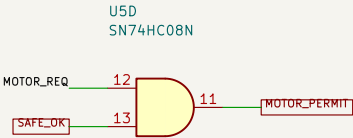
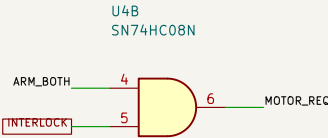
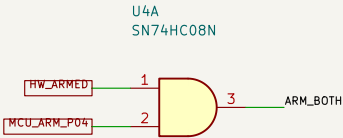


SAFE_N: tylko rezystory i kolektory. RESET latch: SAFE_OK po dwóch bramkach Schmitta.

WD: cel 50-150 ms przy 3V3, do pomiaru. ARM: zwolnić i nacisnąć ponownie po błędzie.

Hardware interlock, watchdog and manual arm; review, hardware not tested		
Prototype / schematic + PCB review		
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File: WD.kicad_sch		
Title: EGRLab P04 SAFE		
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MOTOR_PERMIT = HW_ARMED & MCU_ARM & INTERLOCK & SAFE_OK. PWM_OUT = PWM & MOTOR_PERMIT.



SENSOR_PERMIT = SENSOR_ENABLE & TEST_KEY & INTERLOCK & SAFE_OK. Fizyczny ARM nie jest potrzebny do samego czujnika.

P07 pozostaje HOLD do kontroli rzeczywistego BTS7960. P04 nie ustala prądu ani kierunku mostka.

Hardware interlock, watchdog and manual arm; review, hardware not tested
Prototype / schematic + PCB review
Sheet: /OUT/
File: OUT.kicad_sch

Title: EGRLab P04 SAFE

Size: A3	Date: 2026-09-25	Rev: P04-R1
KiCad E.D.A. 10.0.6		Id: 6/6